

The EverFlavor AI

Multi-Agent Diet & Discovery System

Capstone Project Proposal – Week 3

Assignment: Phase 01 - Assignment Project Proposal (Week 3)

Course: ITAI 2277 - Artificial Intel Resource

Institution: Houston City College

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Main Points

1. The Idea

The Ever Flavor AI is a conversational multi-agent system that helps people create delicious, authentic recipes from cultures around the world without the stress, decision fatigue, or research. Users simply talk to the system in natural language. For example:

“I have 550 calories left, no pork, I’m tired after work, and I want something bold from Middle Eastern or Latin American cuisine. I can drive 15 minutes.”

The system then delivers:

- ☐ A complete, step-by-step recipe from one of five major global cuisine families:
 - Asian cuisine
 - European cuisine
 - Latin American cuisine
 - African cuisine
 - Middle Eastern cuisine
- ☐ Smart substitutions when the user is missing ingredients
- ☐ Accurate calorie and nutrition information
- ☐ The nearest specialty supermarket for that region (Chinese market, Middle Eastern grocery, Latin American store, African market, European deli, etc.)

- ☐ Full respect for every religious, health, medical, and ethical food restriction the user declares

Three Specialized Agents Working Together:

- ☐ **Nutritionist & Preference Agent** – Tracks remaining calories, weekly goals, and all user restrictions
- ☐ **Chef Agent** – Creates authentic recipes and offers practical substitutions
- ☐ **Sourcing Agent** – Finds the closest specialty supermarket within the user's driving radius

No more flipping through outdated cookbooks or trying to remember recipes seen on social media. EverFlavor AI creates everything the user needs in one conversation.

2. The Problem We Are Trying to Solve

Many people feel stuck when it comes to deciding what to eat. They want interesting, flavorful meals from different cultures, but everyday life gets in the way.

This problem is especially common for four groups:

The Busy Professional - Works long hours under high stress. Sees eating mainly as a way to get energy. Plain chicken and rice feels boring, yet they have almost no energy left to cook after work.

The Impatient Foodie - Loves bold, exciting, and gourmet-style flavors. Has a packed schedule and very little free time. Dislikes bland frozen meals or basic home cooking but lacks the hours needed to prepare and clean up complex recipes.

The Overworked Parent - Balances a career, household responsibilities, and childcare. Is tired of repeating the same simple kid-friendly meals. Wants more creative and interesting food for the family but usually settles for whatever is fastest.

The Efficiency-Focused Single or Student - Lives alone or in a shared/dorm space. Feels that cooking for one takes too much time and effort. Quickly gets bored of microwave ramen or fast food, but lacks kitchen confidence or free time to change the routine.

Additional barriers make the problem worse:

- ☐ Not knowing where to buy specialty ingredients for global cuisines
- ☐ Missing ingredients at home and not knowing good substitutions
- ☐ Strict religious rules, food allergies, medical conditions, or ethical choices

- ❑ Decision fatigue from cookbooks and forgotten social-media recipes

EverFlavor AI solves these problems by handling recipe creation, substitutions, nutrition, restrictions, and ingredient sourcing in a single, easy conversation.

3. Data

The system uses reliable, publicly available data sources:

- ❑ **Nutritional data:** Open Food Facts API and USDA FoodData Central for accurate calorie and nutrient information
- ❑ **Location data:** Google Places API to locate real specialty supermarkets by cuisine region
- ❑ **User profile data** (kept session-based in the prototype):
 - Remaining daily/weekly calories
 - Declared food restrictions (religious, medical, ethical)
 - Preferred cuisine families
 - Maximum driving distance
- ❑ **Recipe knowledge:** Generated and adapted by the Chef Agent, then checked against nutritional databases and the safety filter

No large private datasets or custom model training are required.

4. Resources We Need

Technical Resources

- ❑ CrewAI (or similar framework) for multi-agent orchestration
- ❑ Gradio for a simple, interactive chat interface
- ❑ Open Food Facts API + USDA Food Data Central
- ❑ Google Places API (prototype limited to one metro area for scope control)
- ❑ A capable large language model with strong tool-calling ability

Safety & Architecture Resources

- ❑ Structured user profile to store restrictions
- ❑ Hard-coded post-generation filter that scans every ingredient (this filter cannot be overridden by the language model)

Scope Resources

- Focused prototype scope:
 - Five major cuisine families only
 - One metro area for mapping
 - Clear substitution logic
 - Full restriction handling

5. Technical Approach Refinement

- Multi-agent design using CrewAI so each agent has a clear role, memory, and access to tools
- Gradio chat interface for easy live demonstration during class presentations
- Nutrition grounding through public food databases instead of relying only on the language model
- Specialty store search via Google Places API with geographic limits for the prototype
- Smart substitution system built into the Chef Agent so recipes remain usable even when ingredients are missing
- Independent safety layer that runs after recipe generation and cannot be bypassed

The prototype is intentionally limited in geography and cuisine range so the team can deliver a working, reliable system within the course timeline.

6. Orchestration Strategy

EverFlavor AI uses CrewAI to coordinate the three specialized agents in a clear, sequential workflow with shared context.

Process Flow (Sequential)

1. **Nutritionist & Preference Agent** receives the user request first.
 - a. It extracts remaining calories, all declared restrictions (religious, medical, ethical), preferred cuisine families, and driving radius. It builds a structured constraint profile that is passed downstream.
2. **Chef Agent** receives the constraint profile.

- b. It generates an authentic recipe from one of the five allowed cuisine families, applies calorie limits, and produces practical substitutions when ingredients are missing. The output is a complete recipe + substitution list.
- 3. **Sourcing Agent** receives the final ingredient list.
 - c. It queries the Google Places API (geofenced to the prototype metro area) and returns to the nearest specialty supermarket that is likely to carry the regional ingredients.

Key Orchestration Features

- ☐ Shared memory / context so every agent sees the same restrictions and calorie budget.
- ☐ Explicit task dependencies (Chef cannot start until Nutritionist finishes; Sourcing waits for Chef).
- ☐ Tool assignment is role-specific:
 - Nutritionist → nutrition APIs + restriction checker
 - Chef → recipe generation + substitution logic
 - Sourcing → Google Places API
- ☐ Independent safety filter runs after the Chef Agent finishes and before the final answer is shown to the user. This filter is outside the LLM and cannot be overridden.

This sequential + safety-filter design keeps the system simple enough for a 16-week course while still demonstrating real multi-agent collaboration.

7. Human-in-the-Loop (HITL)

Because food recommendations can affect health and religious practice, EverFlavor AI includes deliberate human oversight points:

1. **Safety Gate (Mandatory in Prototype)**
 - a. After the Chef Agent generates a recipe and the automated filter runs, the system can pause and surface the recipe + restriction checklist for a human reviewer (team member or tester) before it is shown to the end user. This is especially important for allergy and religious-restriction cases.
2. **Substitution Review**
 - b. When the Chef Agent proposes substitutions for missing ingredients, the user (or a reviewer) can accept, reject, or request alternatives in the chat interface.

3. Preference Feedback Loop

- c. After a meal suggestion is delivered, the user can give simple feedback (“too spicy”, “perfect”, “not authentic enough”). In later phases this feedback can be stored to improve future recommendations.

4. Override Capability

- d. The user always retains final control. They can ask the system to regenerate the recipe with stricter or looser constraints at any time.

In the Week 3–6 prototype we will implement the safety gate and basic user feedback. Full automatic learning from feedback is planned for later milestones if time allows.

CrewAI’s built-in human-feedback / human-in-the-loop mechanisms (Flow-based or webhook-based) will be used so the pause-and-review step is native to the orchestration rather than bolted on.

8. Timeline and Milestone Planning (16-Week Course)

Week	Focus	Key Deliverables
1	Team formation & orientation	Team roles assigned, tools set up (GitHub, shared Drive, etc.)
2	Problem & scope finalization	Final problem statement, user personas locked, feasibility check
3	Project Proposal (Current)	Full proposal submitted + presentation
4	Environment & basic agents	Python environment, CrewAI installed, simple Nutritionist Agent working
5	Chef Agent + Gradio skeleton	Chef Agent generates basic recipes, first Gradio chat interface
6	Restrictions & safety filter	Hard-coded allergen/religious filter working, integrated with Chef
7	Nutrition grounding	Connect Open Food Facts + USDA APIs, calorie calculation accurate
8	Sourcing Agent	Google Places API integrated (limited to one metro area, e.g. Houston)

9	Full sequential orchestration	All 3 agents working together in correct order + shared context
10	Human-in-the-Loop + substitutions	Safety gate + user can accept/reject substitutions in Gradio
11	Testing & metrics	Measure the 5 success metrics, fix major bugs
12	Polish & edge cases	Better prompts, error handling, improved UI in Gradio
13	Optional: Computer Vision stretch	Photo upload → basic “does this look like the recipe?” check (if time)
14	Documentation & demo prep	User guide, architecture diagram, presentation slides
15	Final testing & peer feedback	Full end-to-end demos, collect feedback
16	Final presentation & submission	Live Gradio demo + final report

Development Notes

- ☐ Core system remains a Python + Gradio application.
- ☐ Computer vision is an optional stretch goal scheduled in Week 13.
- ☐ Human-in-the-Loop and full sequential orchestration will be functional by Week 10.

9. Success Metrics Definition

1. Nutritional Accuracy

- a. The system’s calorie calculations must stay within a 5% margin of error compared to careful manual calculations using the same data sources.

2. Restriction Compliance

- b. 100% of recipes shown to the user must pass the automated filter for every declared religious, medical, or ethical restriction.

3. Sourcing Accuracy

- c. The Sourcing Agent must return only real specialty stores. No hallucinated store names or addresses are acceptable.

4. Substitution Quality

- d. When ingredients are missing, the suggested substitutions must keep the dish recognizable and satisfying.

5. Practical Usability

- e. For realistic queries from busy professionals, impatient foodies, overworked parents, and efficiency-focused singles/students, the system must deliver a complete, usable answer in a single conversational turn.

10. Risk Assessment

Risk	Impact	Likelihood	Mitigation Strategy
LLM forgets a food restriction	High	Medium	Hard-coded post-generation filter that cannot be overridden
Hallucinated store locations	Medium	Medium	Strict use of Google Places API + result validation
Inaccurate calorie counts	Medium	Low–Medium	All calculations grounded in Open Food Facts / USDA data
Scope creep	High	Medium	Strict limits: 5 cuisine families + one metro area
API rate limits or unexpected costs	Low–Medium	Low	Use free tiers, caching, and limited demo queries
Poor quality substitutions	Medium	Medium	Test and refine substitution logic with clear criteria

11. Ethical Considerations

Primary Ethical Concern

Recommending food that violates a user’s religious beliefs, triggers an allergy, or conflicts with a medical condition can cause real harm.

How We Address It

- ☐ All restrictions are stored in a structured user profile (not only in chat history)
- ☐ After the Chef Agent creates a recipe, an independent filter automatically checks every ingredient

- ☐ Any conflict causes the recipe to be rejected or rewritten before the user sees it
- ☐ Safety is never left only to the language model's memory or prompt instructions

Restrictions Fully Supported by the Safety Filter

Religious

- ☐ Judaism (Kosher): No pork or shellfish; meat and dairy not eaten together
- ☐ Islam (Halal): No pork or alcohol; meat prepared according to Islamic rules
- ☐ Hinduism: Many avoid beef; many follow vegetarian diets
- ☐ Buddhism: Many avoid meat
- ☐ Jainism: Strict vegetarianism; avoids root vegetables (potatoes, onions, garlic, etc.)
- ☐ Christianity (Catholicism): Some avoid meat on Fridays during Lent

Health & Medical

- ☐ Food allergies (peanuts, tree nuts, milk, eggs, shellfish, etc.)
- ☐ Celiac disease (gluten-free)
- ☐ Lactose intolerance
- ☐ Diabetes (controlled sugar and refined carbohydrates)

Ethical & Philosophical

- ☐ Vegetarianism
- ☐ Veganism

Additional Ethical Practices

- ☐ Clear disclaimer that EverFlavor AI is a helpful planning tool, not medical or religious advice
- ☐ Location data is used only for the current session in the prototype
- ☐ Calorie calculations are transparent so users can verify the numbers

12. Ingredient Interaction Knowledge

Knowing what to combine, and what never to combine, is a distinct capability from generating a recipe. We separate it into three tiers because they fail differently and therefore need different mechanisms. Two tiers are in prototype scope. The third is a planned extension.

Tier 1: Hard Constraints (Never Learned)

Some combinations are unsafe rather than merely unpleasant, and no learned component may ever be in a position to decide they are acceptable. These live in the same non-overridable filter layer as the allergen checks.

Examples include:

- ❑ Cross-contamination between raw poultry or ground meat and ready-to-eat items in the same preparation step
- ❑ Prolonged contact between heavily acidic or salty foods and reactive cookware such as untreated aluminum
- ❑ Minimum internal temperature floors for poultry, ground meat, and eggs, which no generated instruction may contradict

This tier is **in scope for the prototype** and is implemented as hard rules, not as model behavior.

Tier 2: Chemistry and Physics (Encoded as Rules)

These combinations are not dangerous; they simply fail in predictable ways. Because they are predictable, a rule base handles them more reliably than a learned model. The Chef Agent consults this rule set before finalizing a recipe.

Examples include:

- ❑ Raw pineapple, papaya, kiwi, and fresh figs contain proteases that prevent gelatin from setting and can turn marinated proteins mushy
- ❑ Acid added directly to dairy curdles it (cream must be tempered into an acidic sauce)
- ❑ Salt draws water from vegetables by osmosis (intentional when salting eggplant in advance, a mistake when salting a stir-fry early)
- ❑ Baking soda requires an acid in the formula, while baking powder supplies its own
- ❑ Cornstarch breaks down under sustained heat, while a flour roux holds
- ❑ Maillard browning requires a dry surface above roughly 140°C; wet protein steams instead of searing

An important property of this tier is that almost every rule is **conditional** rather than absolute. For example, raw pineapple blocks gelatin, but canned pineapple does not, because heat denatures the enzyme. Encoding the condition explicitly is more reliable than hoping a model will infer it. This tier is **in scope** as a lookup consulted during recipe finalization.

Tier 3: Flavor Compatibility (Learnable – Planned Extension)

This is the tier where a learned model earns its place. It is a planned extension rather than prototype work.

Two signals are available:

- Published food pairing research that analyzes shared volatile aroma compounds between ingredients (CulinaryDB already listed in our sources comes from the same research group)
- Co-occurrence mining from a large recipe corpus, which produces embeddings where culturally coherent combinations cluster

The direct application is to our existing substitution logic. Rather than relying only on a hand-written table, the system would learn substitutability from context and use it to rank candidate substitutions.

Bias risk we must manage: Co-occurrence learns what is common, not what is good. A recipe corpus weighted toward Western sources encodes Western norms as if they were universal. A model trained on such a corpus would quietly score authentic combinations as errors (e.g., fish sauce with caramel, cinnamon in savory Middle Eastern meat dishes, chocolate with chile). If this tier is built, the corpus must be stratified by cuisine and the model evaluated per cuisine rather than reported as a single global accuracy figure.

Design principle: Tiers 1 and 2 may reject a recipe. Tier 3 may only reorder suggestions.

Allowing a learned flavor model to block anything is precisely how a system ends up rejecting a correct recipe from an underrepresented cuisine.

13. Long-Term Extension: Robotic Cooking

The furthest extension of this project is a physical system that cooks rather than instructs. We describe it here to show where the architecture leads, while being clear that it is well beyond a 16-week course and that only narrow slices of it are prototypable.

Manipulation vs. Timing

The common assumption is that the robotic arm is the central challenge. It is difficult, but it is not where the meaningful AI work sits. Food is deformable, variable, and slippery. Timing is the real modeling problem. Cooking is a scheduling problem under resource constraints with irreversible actions. Limited burners, limited pans, limited hands. The binding constraint is that

everything should finish hot at the same moment. This is a temporal planning problem, and one where a system can be reliably better than a home cook.

Stove Understanding as System Identification

Gas and electric are not cosmetic variations; they are different control problems. Gas responds almost immediately. Electric coil carries substantial thermal lag (30–90 seconds of stored heat). Induction responds quickly but heats only ferrous cookware. The approach is to learn a thermal response model for each hob and pan combination using standard system identification techniques.

Food Chemistry Belongs in the Rule Base

Marinating, brining, and most doneness reasoning do not require a learned model. These belong in the Tier 2 rule base and feed the scheduler. Learning genuinely helps in one place: doneness estimation from vision.

Prototypable Slices (in order of increasing hardware risk)

1. **Coaching mode** (no actuation): A camera and temperature probe observe a human cook and advise when to flip, when the pan is too cold, and when to add the next ingredient.
2. **Instrumented single hob**: Thermocouple or infrared sensor + camera + controlled induction hob that holds a target temperature through disturbances.
3. **One degree of manipulation freedom**: A single rotating stirring arm in a pot (useful for risotto, roux, and anything prone to scorching).

Safety Principle

A system that undercooks poultry is a pathogen delivery mechanism. Any system operating near hot oil carries fire risk. Every version requires hard temperature floors and hard thermal limits that no learned policy may override. This is the same architectural principle as the restriction filter in the current prototype.

The Bridge from the Current System

Our present system outputs a recipe (an unordered list of instructions). A robotic system requires a temporal plan a schedule of parallel tasks with dependencies and durations. Converting recipe text into a structured task graph is the bridge between the two. That conversion is useful on its own, even with no robot involved, because it would let the current system tell a user what to start

first and what to do while waiting. It is the one piece of this extension that could be prototyped within the course timeline if the schedule allows.

14. Summary

The EverFlavor AI combines multi-agent intelligence, global cuisine knowledge, practical substitutions, specialty-store finding, and strong safety architecture into one useful system. It is designed for real people with real time constraints and real dietary needs, while remaining realistic to build and demonstrate within a 16-week course.

15. Sources

1. Nutritional Data (Calories & Nutrients)

- USDA FoodData Central (official U.S. government) Website + free API:
 - <https://fdc.nal.usda.gov/>
- Open Food Facts (global, crowd-sourced) API documentation:
 - <https://openfoodfacts.github.io/documentation/>

2. Recipe & International Cuisine Data

- Food.com Recipes and User Interactions (Kaggle)
 - <https://www.kaggle.com/datasets/shuyangli94/food-com-recipes-and-user-interactions>
- RecipeDB – 118,000+ recipes from 74 countries
 - <https://cosylab.iiitd.edu.in/recipeadb>
- CulinaryDB – Structured recipes across 22 world regions
 - <https://cosylab.iiitd.edu.in/culinaryadb/>
- Hugging Face – Recipes with Nutrition
 - <https://huggingface.co/datasets/datahiveai/recipes-with-nutrition>